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The multifunctional artificial reef and its role in the defence of the Mediterranean coast

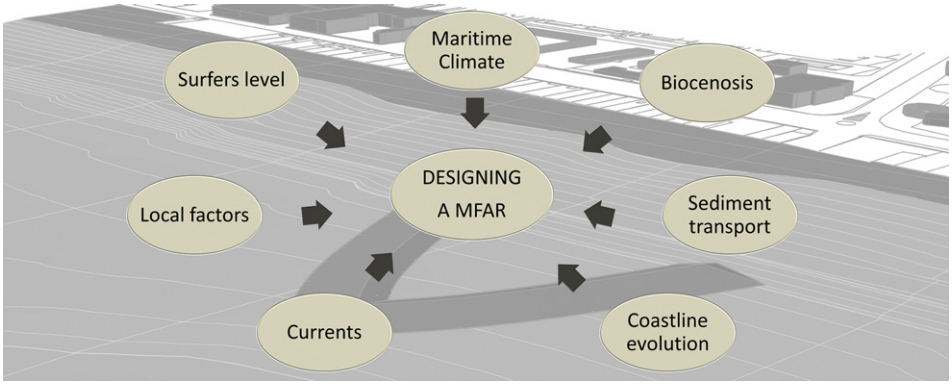
I. López [a,*], H. Tinoco [a,b], L. Aragonés [a], J. García-Barba [a]

[a] Dept. of Civil Engineering, University of Alicante, Carretera San Vicent del Raspeig s/n, 03690 Alicante, Spain
[b] OHL Croatian Branch, Šubićeva 20, 10000 Zagreb, Croatia

HIGHLIGHTS

- Multipurpose surfing reef in the Mediterranean Sea
- Alternative shore defence system
- Erosion problem in Babilonia beach, Guardamar, Spain
- Sand accretion in the shore
- Improving biocenosis in the study area

GRAPHICAL ABSTRACT



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ABSTRACT

Multifunctional artificial reefs (MFAR) are being implemented around the world, due to their ability to provide an environment where a sports–economic–recreational use and environmental improvement is implemented, and are also elements of coastal defence. However, a lot of failures have been recorded, possibly due to disregarded local factors in the formulations used, and there is no method that has encompassed all these factors, in order to take them into account in its design. The aim of this paper was to provide the coastal engineer with a method that would be used for the design of such reefs. To do this, the Babilonia beach of Guardamar del Segura, Alicante (Spain), was chosen because it is a fully anthropised area (with houses in the Maritime-Terrestrial Public Domain, marina, channelling and river mouth) with continuous regression, in which all the elements considered in this study, were treated. Based on the performance obtained in studies and projects worldwide, the climatic characteristics, biocenosis, sediment transport, settlements and liquefaction and the evolution of the coastline, were analysed. The multidisciplinary study carried out showed that with the implementation of a MFAR, the problem was reversed. Furthermore, the area was provided with a playful-economic use, and could be used 60% of the time, by surfers whose skill level were low to intermediate, without forgetting that the diversity of the marine ecosystem in the area was increased.

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1. Introduction

Erosion is a problem that affects most of the world's beaches. The measure most commonly used to curtail it is the hard-engineering

* Corresponding author.
E-mail address: isalopu.il@gmail.com (I. López).

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Table 1
Collection on the multifunctional artificial reef.

MFAR	Goal	Type	Structure	Modelling	Surfability study	Peel angle	Currents	Local factors	Coast evolution	Distance to the coast	Erosion/accretion	Surfer level	Proliferation wildlife	Surf improvement	Reference
Bargara (Australia, 1997)	Surf	Rock	Parallel to the coast	X	X	X	X	X	X	X	X	X	X	✓	1
Cables (Australia, 1998–1999)	Surf	Rock	Boomerang	Numeric and scale	✓	✓	✓	Waves and tides	YES (after construction)	300 m	No change	X	X	✓	2–7
Narrowneck (Australia, 2000)	Coastal defence and surf	SFGC[a] without seam	V slit in the middle	Numeric	✓	✓	✓	Waves and tides	✓	250 m	Accretion 50 m	3–7	✓	YES (but under large waves)	8–11
Prattes (USA, 2001)	Surf	SFGC[a] without seam	Delta	Numeric	X	X	X	Waves and tides	YES (after construction)	80 m	No change	X	✓	NO (removed 2008X2010)	12
Mount Maunganui (NZ, 2006)	Surf	SFGC[a] without seam	V	Numeric and scale	✓	X	✓	Waves, tides and sediment flow	YES (after construction)	250 m	Accretion	X	✓	✓	13
Opunake (NZ, 2010)	Surf	SFGC[a] without seam	Parallel to the coast	Numeric	✓	✓	X	Waves and tides	X	X	X	X	✓	X	14–15
Boscombe (UK, 2009)	Coastal defence, economy and surf	SFGC[a] without seam	Perpendicular to the coast	Numeric and scale	✓	✓	✓	Waves and tides	YES (after construction)	225 m	No change	3–6	X	NO (extremely complex)	16
Kovalam (India, 2010)	Coastal defence, economy and surf	SFGC[a] without seam	Oblique to the coast	Numeric and physical	X	X	✓	Waves and tides	X	X	Accretion	X	✓	NO (few days of decent surf)	17
Ventura Oil (USA, unbuilt)	Coastal defence and surf	SFGC[a] without seam	L	Numeric and scale	✓	✓	✓	Waves, tides and sediment flow	X	250 m	X	4–7	X	X	18
Long Branch (USA, unbuilt)	Surf	SFGC[a] without seam	V	Numeric	X	X	✓	X	X	X	X	X	X	X	19
Newquay (UK, unbuilt)	Surf	SFGC[a], or SFGC[a] with rocky core, or SFGC[a] with pneumatics core	V slit in the middle	Numeric	X	X	✓	X	X	X	X	X	X	X	20–21
São Pedro (Portugal, unbuilt)	Surf	SFGC[a]	Parallel to the coast	Numeric	X	✓	✓	Waves, tides and winds	X	X	X	X	X	X	22–23
Orewa (NZ, unbuilt)	Coastal defence and surf	SFGC[a] without seam	Four sets of L reefs	Numeric	X	X	✓	Waves, tides and sediment transport	X	X	X	X	X	X	24
Barcelona (Spain, unbuilt)	Coastal defence and surf	SFGC[a] without seam	V	Numeric	X	X	✓	X	✓	180 m	X	X	X	X	25
Brevard (USA, unbuilt)	Coastal defence and surf	SFGC[a] without seam	V	Numeric	X	X	✓	X	X	300 m	X	X	X	X	26

1) Jackson and Corbett (2007), 2) Bancroft (1999), 3) Button (1991), 4) Hurst (1996), 5) Lyons (1992), 6) Pattiaratchi (1999), 7) Pattiaratchi (2003), 8) Black and Mead (2001a), 9) Hutt et al. (1998), 10) Black et al. (1998), 11) Jackson et al. (2007), 12) Borrero and Nelsen (2003), 13) ASR (2008b), 14) Taranki (2011), 15) Taranki (2009), 16) Mead et al. (2010), 17) Frazerhurst (2008), 18) Schlosser (2005), 19) ASR (2013a), 20) Challinor and Weight (2005), 21) ASR (2002), 22) Bicudo and Cardoso (2012), 23) Cardoso et al. (2012), 24) ASR (2013b), 25) García Márquez (2009) and 26) ASR (2008a).

[a] SFGC: Sand Filled Geotextile Containers.

method (Ng et al., 2014). The usual coastal defence solutions are based on the implementation of rigid elements, such as breakwaters or dikes. Breakwaters and dikes are man-made elements that cause great environmental and aesthetic impacts (De la Peña, 2007) on the beach. These elements work by blocking the energy reaching the beach, and they produce diffraction around the edges that favour the formation of a hemi-tombolo or over time a tombolo, if sufficient solid littoral transport is available (De la Peña, 2007).

While the hard-engineering solution is likely to remain valuable and viable for coastal protection, in the global option, there has been a growing trend in the application of a soft-engineering solution known as the multifunctional artificial reef (MFAR) (Ng et al., 2015). These submerged structures, have the potentials of providing coastal protection and improving recreational facilities, such as the beach width, and surfing respectively (Mead and Black, 1999). Also, they reduce the impact of waves on the shore, through dissipation (by breaking the waves), and/or rotation (decreased by the realignment of the waves of longshore currents) (Black and Mead, 2001b; Mead and Black, 1999). On the other hand, the structures introduce a diversified ecosystem, creating areas of natural marine restocking, as they cause the proliferation and growth of living organisms, and the development of algae and marine fauna (Romero, 1998).

MFAR, commonly known as multifunctional reefs or artificial surfing reef, is a relatively young soft-engineering approach, with the first small-scale action of Burkitt Reef (Queensland, Australia) undertaken in 1997, the large-scale installation of the Cable Station Reef (Perth, Australia) in 1998, and the newest Reef Borth (Wales, UK) carried out in 2012 (Lokesha et al., 2013).

The overall performance of MFAR is difficult to assess, due to the limited number of reefs built within its few years of operation, the lack of programs and post-monitoring data, and the lack of specialized magazines (Ng et al., 2015). As with most new technologies, mistakes have been made and we can learn from them. Ng et al. (2013) summarized the main features and performance of the MFAR existing at that time, and presented a set of design criteria derived from the "best" MFAR. An important aspect is to pay attention to the geometry of these artificial reefs and their location in relation to the shoreline, as there are many cases of erosion resulting from the implementation of submerged dikes, which are wrongly sized and located. Examples were collected from Ranasinghe and Turner (2006). Another important aspect to consider are settlements (and in some cases liquefaction) that have occurred in structures located on sandy bottoms causing major failures (Medina et al., 2006; Sumer and Fredsøe, 2002).

The different cases and studies can be differentiated, in relation to the materials used for its implementation, the design methodology used, the ability to proliferate the flora and fauna in the beach, the main execution of its lens, and the performance thereof. Many of these aspects are set out in Table 1.

Therefore, the objective of this work has not been widely regenerate a degraded area, but above all make a timely action in this area in which to study the maximum number of variables to generate a method of study of artificial reefs multipurpose and serve as a user-friendly tool for coastal engineer. To do this, a multidisciplinary study, which took into account the morphology of the beach, the biocenosis, waves, sediment transport, geotechnical aspects (settlements and liquefaction) and the sports–economic–playful aspect of this type of reef, was carried out. All this was achieved, by taking into account the local aspect of the formulas used, in order to avoid design errors.

Based on the experience gained from previous studies, a MFAR was designed, and it was located south of the River Segura in the town of Guardamar del Segura (Alicante, Spain), with a high degree of regression in its shoreline and fully anthropised to be used as an example, with the construction of houses within the terrestrial maritime public domain (DPMT), marina, channelling and river mouth.

2. Study area

The study area was located Southeast of Spain, particularly south of the Segura River in the town of Guardamar del Segura (Fig. 1). It was an open beach, which is currently suffering from serious beach erosion problem (Aldeguer Sánchez, 2008). It was a highly urbanized area with different anthropic elements, among which were the channelling

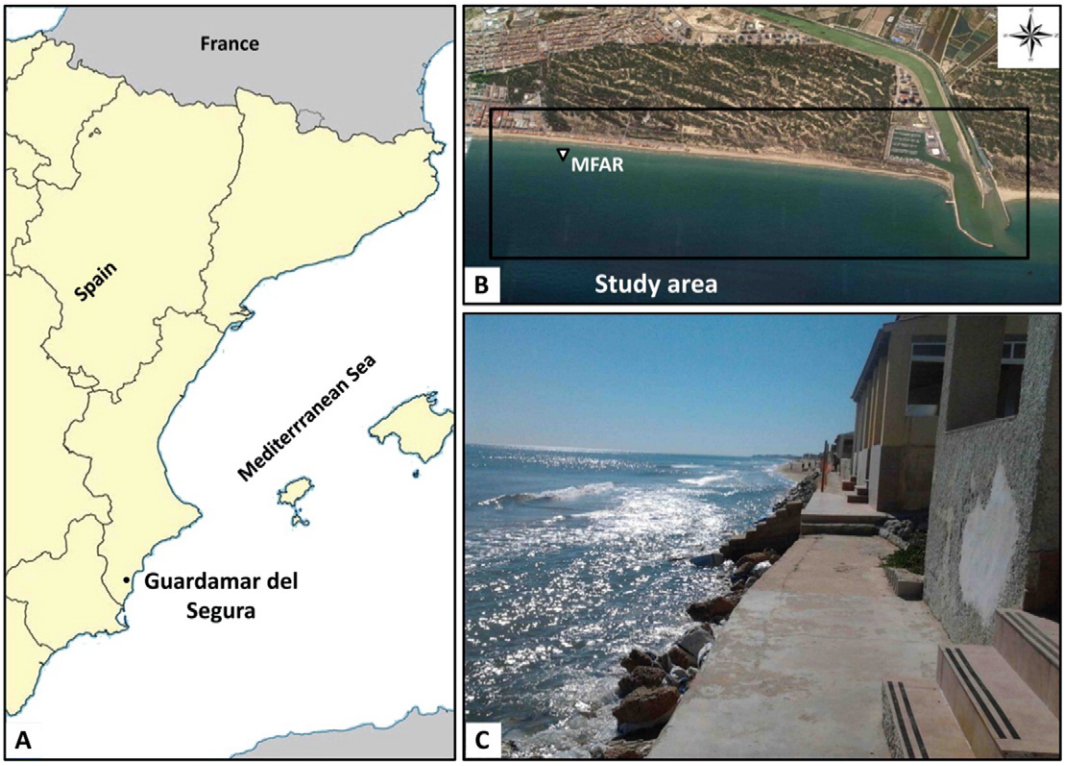


Fig. 1. A) Location of Guardamar del Segura in Spain. B) Location of the study area and location of the reef in Guardamar del Segura. C) Detail the current state of Babilonia beach.

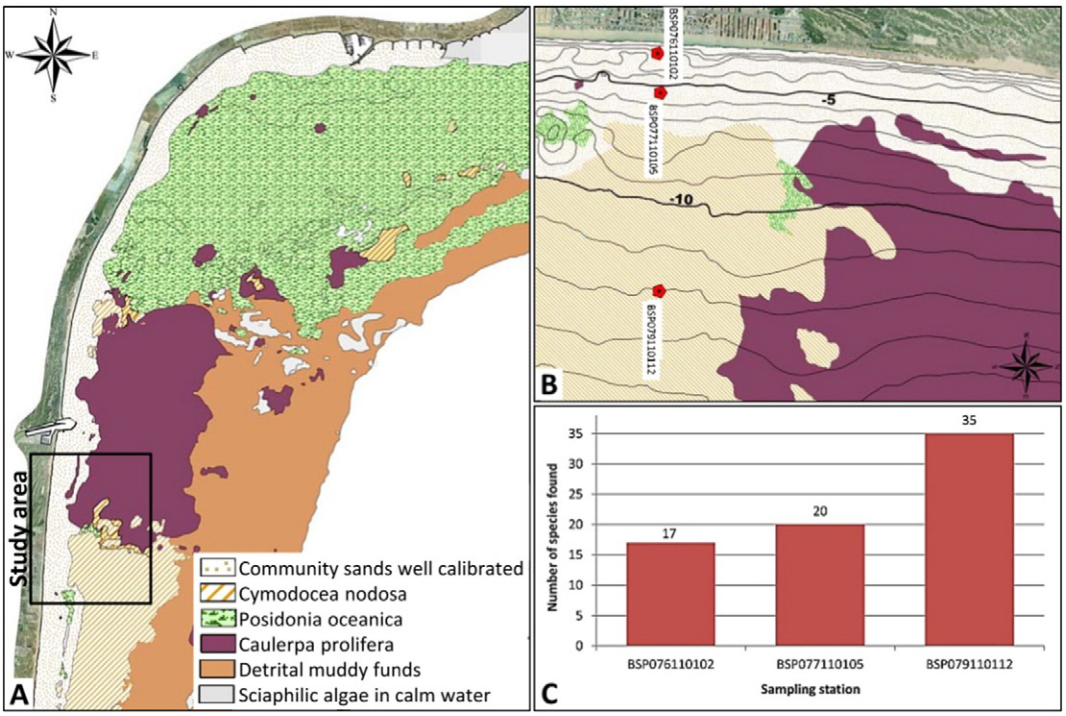


Fig. 2. A) Bionomic mapping of the area. B) Details of the study area and location of sampling stations. C) Number of species found in each of the seasons.

of the Segura River built in 1990, a marina built within the river mouth between 1996 and 1998 (Fig. 2B), and various dams that prevent sediment supply (Aldeguer Sánchez, 2008; Aragonés et al., 2016).

On the Babilonia beach, there was a line of houses built in 1948, which were now a few meters from the shoreline (Torres, 1997), and during storms, waves reflected on the walls of breakwaters built in front of the houses by the owners, to protect them (Fig. 1C). To alleviate this problem, and to protect and improve the marine biocenosis in the area, a multifunctional artificial reef which was located in the coordinates 38°05′24″N and 0°38′42″W, was designed (Fig. 1B). Furthermore, it aimed to eliminate the breakwaters at the river mouth, in order to allow longshore sediment transportation (Aldeguer Sánchez, 2008).

The tides in the Mediterranean Sea were not constant over time, and the types of tide (diurnal, semidiurnal or mixed), varied. In addition, the tides were conditioned by various weather factors, long waves, or other factors that distorted the purely astronomical wave and caused a change in their amplitudes, to a greater extent than expected. In the study area, astronomical tides had amplitudes between 20 cm and 30 cm, but sometimes, the impact of local factors could cause the amplitudes of the tides to rise to one metre (ECOLEVANTE, 2006).

2.1. The biocenosis of the marine environment

The biocenosis of the seabed in the study area was obtained from ECOLEVANTE (2006), the analysis of the data set revealed that the study area has a large number of identifiable bionomic aspects. In the study, a number of experimental plots (25 × 25 m) were delimited, in the places in which it took place; i) visual census in the pelagic domain (nekton). ii) Visual tours in each of the areas, for the identification and quantification of other macrofloristic and macrofaunal elements. The number of transects in each parcel was not less than five, thus, performing a visual census of all plots, was assured.

The distribution of all communities is shown in Fig. 2. As can be seen, the places studied, were characterized by great diversity, because of the presence of different bionomic communities, and the wide variety of structural adjustments. Among the highlighted bionomic communities, we identified the community of well calibrated fine sand, prairie *Cymodocea nodosa*, prairie *Caulerpa prolifera*, and some patches of *Posidonia oceanica*.

P. oceanica is a community with little representation throughout the study area, but not in the north of it (ECOLEVANTE, 2006), probably due to the unfavourable conditions that cause mud to be carried by the river, during the continuous flooding suffered in the study area (Aragonés et al., 2016). After the cessation of contributions of sediments as a result of the canalization of the river, as well as innumerable dams and weirs constructed, the growth of *C. nodosa* and *C. prolifera* has been shown (Aragonés et al., 2016). These two species are able to generate a prolific ecosystem in their surroundings, but it is less than *P. oceanica*, because, it is simpler and less diverse (ECOLEVANTE, 2006).

3. Methodology and results

Here, the process used in the design of the MFAR is described. The elements analysed were: i) The maritime climate in the area, ii) the historical evolution of the coastline, iii) MFAR sizing and skill level of the surfers, iv) modelling environment considering the incorporation of MFAR, v) SMC coastal modelling in which the skill levels of the surfers were checked, vi) settlements and stress behaviour of the soil (liquefaction) and vii) the reef structure, constructive process and cost analysis.

Table 2
Results of the climate study.

Direction	Probability	$H_{s,12}$	T_P
N	0.010	1.37	12.30
NNE	0.017	1.46	12.30
NE	0.070	3.02	13.50
ENE	0.156	3.30	12.30
E	0.249	2.80	11.20
ESE	0.121	1.22	10.10
SE	0.047	1.00	10.10
SSE	0.046	1.00	10.10
S	0.100	1.60	10.10

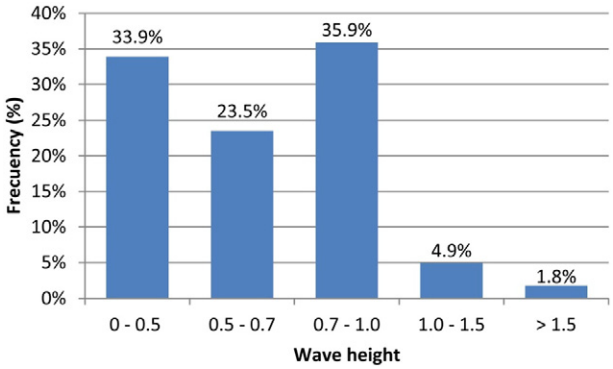


Fig. 3. Frequency of wave height.

3.1. Maritime climate

The incident wave from the database of the SIMAR network provided by the Organismo Público Puertos del Estado, was characterized. These data provided information on the wave height, period, wave direction, and the probability of occurrence from 1958 to 2014. Specifically, the SIMAR 2044025 (38.13°N; 0.50°W), was used.

Thus, the wave data from the SIMAR network were statistically analysed by the program Carol v1.0 (Universidad de Cantabria), and, the wave height $H_{s,12}$ (likelihood of 0.137%), their associated period and the probability of occurrence for each of the incident wave directions, was obtained (Table 2).

From the data of Table 2, the mean flow direction in the study area was calculated using Eq. (1). The mean flow in the region had a direction of 86.6° with respect to north.

$$\text{Direction of the mean flow} = \arctan\frac{\sum p_i H_i \operatorname{sen}\varphi_i}{\sum p_i H_i \cos\varphi_i} \quad (1)$$

Finally, the frequency with which each wave height occurred was obtained (Fig. 3).

3.2. Historical development

The analysis of the evolution of the coastline was carried out, by superimposing a series of georeferenced orthophotos (Ojeda et al., 2013), ranging from the years 1956 to 2014. Specifically, orthophotos from 1956, 1981, 1992, 1994, 1996, 2002, 2005, 2009, 2012 and 2014 were available, thereby, allowing us to observe the trend of the coastline. These orthophotos were supplied by Dirección General de Costas de Alicante y el Instituto Cartográfico Valenciano (Generalitat Valenciana).

Using a GIS system, measurements of the surfaces and the width of the dry beach for each orthophoto was obtained, and the variation of the coastline, and the same trend were analysed.

Thus, it was discovered that the shoreline south of the river Segura swung in an anti-clockwise direction, in line with the results of the mean flow (Fig. 4). This resulted in a regression, which is shown in Fig. 5, for the period of 1956 and 2009. This continuous decline of the coastline was aggravated by the presence of houses within the area of influence of the waves. Fig. 4, shows the intersection of the line of houses and the shoreline. Moreover, an important beach width difference in the north was observed, and therefore, the distance of the shoreline to the houses in 1956, was to a great extent, larger than the distance in 2009 (Fig. 5). In 1956, the beach width was about 41 m, in 1992 it was 31 m, and 15 m in 2005.

The loss of beach width was critical in the northern area of the houses, as shown in Fig. 5, where the sharp decline from the beach coastline is shown over the years to 2005. The annual erosion between 1956 and 1992 was −0.13 m while between 1992 and 2009, the rate of erosion was −0.54 m/year, which was due to the construction of the breakwater of the Segura River in 1992, and the reflection produced in the area, due to the temporary reach of the waves, to the houses. In the last period studied (2009–2014), the beach had lost approximately 11,297 m^2, losing in the last two years an average of 3.57 m/year.

Meanwhile, the northern part of the Segura River was seen, as the beach has being more or less stable over the years, resulting in a build-up of materials along the breakwater at the mouth (Fig. 6). Since the construction of the north breakwater up until now, material accumulation has reached 42 m.

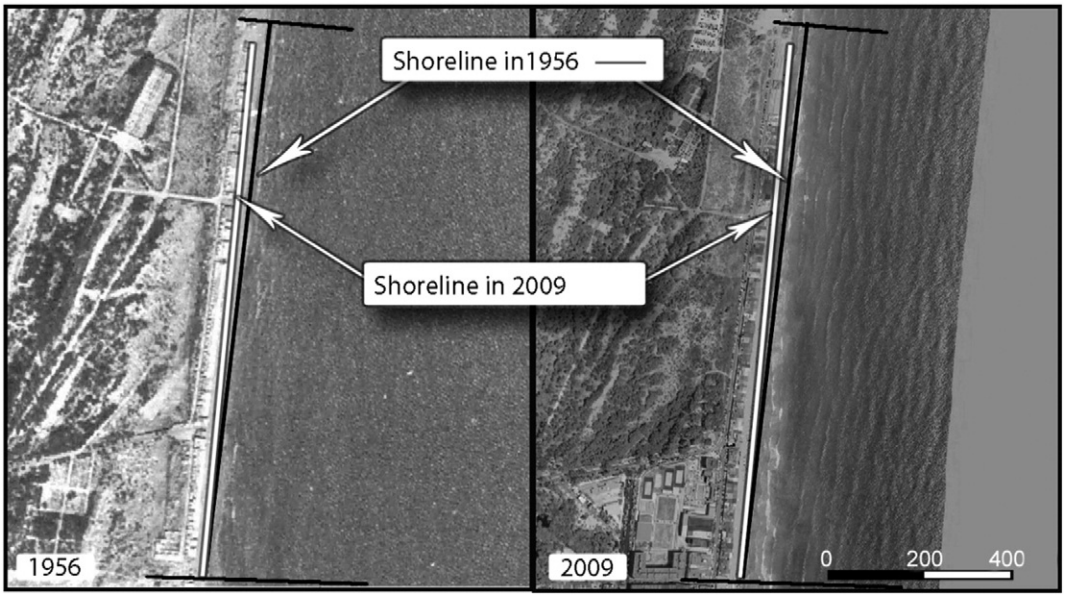
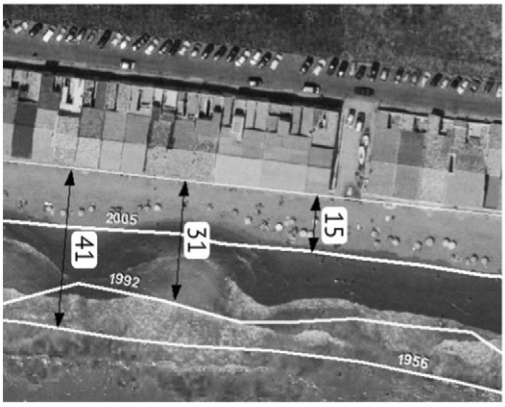


Fig. 4. Ortophotos of the North extreme of Babilonia beach. 1956 (left) and 2009 (right).

A



B

Study period	Area variation (m^2)	Annual regression (m/year)
1956-1981	-3979.9	-0.14
1981-1992	-1113.9	-0.20
1992-1994	-689.3	-0.31
1994-1996	-5081.3	-2.31
1996-2002	-2574.0	-0.39
2002-2005	-934.8	-0.28
2005-2009	-800.3	-0.18
2009-2012	-3443.7	-1.04
2012-2014	-7854.0	-3.57

Fig. 5. A) Evolution of the coastline on the Babilonia beach (1956, 1992 and 2005). B) Table showing the variation of the surface of the beaches and the annual regression between periods.

3.3. Sizing of the MFAR

For the location and sizing of the artificial reef, the first accretions that we wanted to get, and the length of beach that was treated, were defined. The latter was established so as to act on the most affected by the regression in front of the houses (270 m), and the width of the projection was set at about 30 m, so that the width of the beach in 1956 (41 m), could at least be met.

To date, there have been little amount of researchers (Black and Andrews, 2001a; Douglass and Weggel, 1987; Groenewoud et al., 1996; Silvester and Hsu, 1997), who have studied the response of the coast, to the implantation of an artificial reef or submerged dike. However, due to the large number of submerged dikes used for coastal defence (Delaware Bay (USA), Keino-Matsubara Beach (Japan), Niigata (Japan), Lido di Ostia (Italy), Lido di Dante (Italy) Marche (Italy), Palm Beach (FL, USA), Vero Beach (FL, USA), Gold Coast (Australia), and so on), we have been able to establish relationships between the element of defence, and its effect on the shoreline. Among these studies highlighted, the study of Silvester and Hsu (1997) and that of Black and Andrews (2001a), were compared by Ranasinghe and Turner (2006), and they found that they were very similar. Black and Andrews (2001a, 2001b) obtained a formula (Eqs. (2)–(6)), to determine the distance of the reef to the coast (S), and the width (B) thereof. Also, Black and Andrews (2001a) established that for the reef to form a protrusion or tombolo, the value of B/S had to be between 0.1 and 2. On the other hand, Ranasinghe et al. (2010) also proposed a formula based on tests carried out on a reef tank, and the delimiting situations where it would generate erosion or accretion depending on its size and location relative to the coast, were considered (Eq. (7)). This equation establishes the main geometric and placement of artificial reefs relationships, to generate four cell currents, which are needed to produce a certain accretion on the shoreline.

$$\frac{X_{off}}{B} = 0.50 \cdot \left(\frac{B}{S}\right)^{-1.27} \tag{2}$$

$$Y_{off} = S - X_{off} \tag{3}$$

$$x = x' \cdot Y_{off} \tag{4}$$

$$y = y' \cdot Y_{off} \tag{5}$$

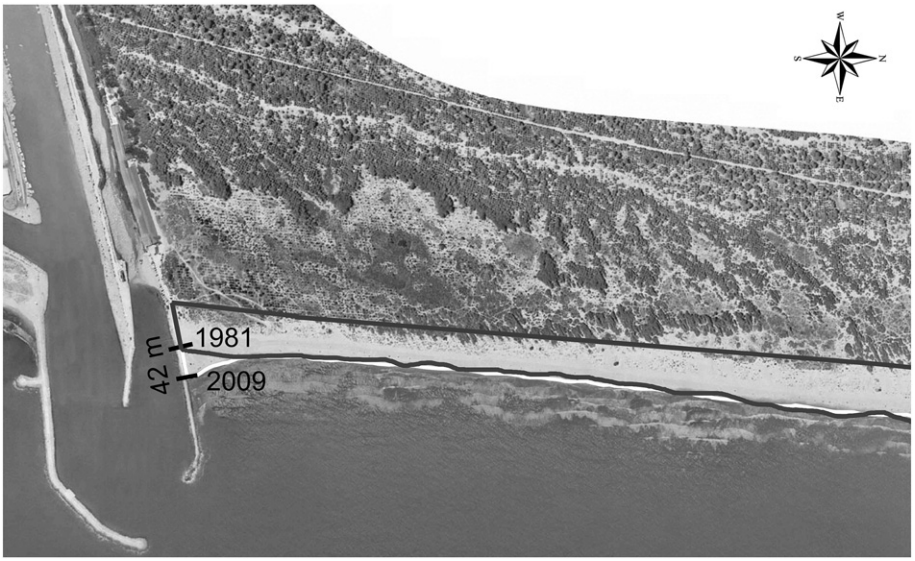


Fig. 6. Evolution of the coastline on the beach north of the Segura River (1981 and 2009).

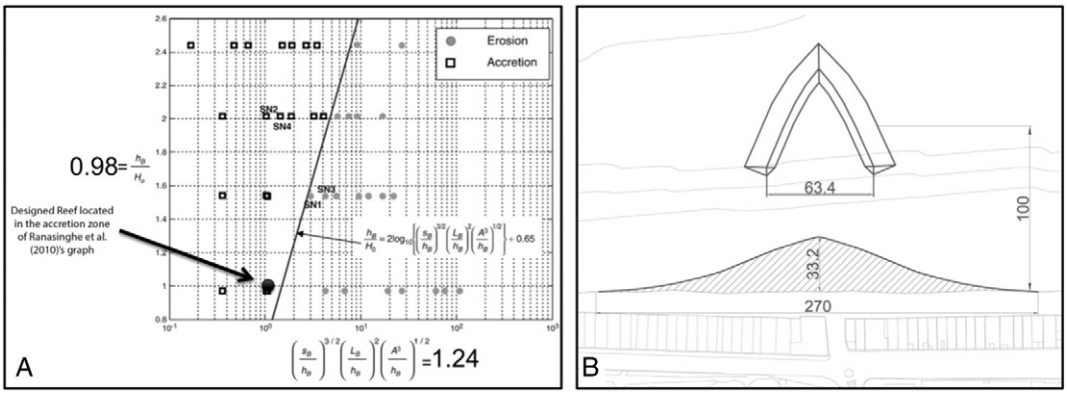


Fig. 7. A) Graph of Ranasinghe et al. (2010) applied to the study reef. B) Simulation of accretion due to the implementation of MFAR.

$$y' = -0.052 + \frac{1.187}{\left\{1 + e^{\left[-\frac{\left(x' - 0.606 \ln\left(2^{1/0.606} - 1\right) - 2.649\right)}{0.606}\right]}\right\}^{0.606}} \quad (6)$$

where:

X_{off} → distance from the vertex of the ledge to the MFAR
Y_{off} → distance from the vertex of the ledge to the original waterline
B → multipurpose artificial reef width which corresponds to the length of a submerged dike
S → distance from the MFAR to the waterline.

$$\frac{h_b}{H_o} = 2\log_{10}\left\{\left(\frac{s_B}{h_b}\right)^{3/2}\left(\frac{L_B}{h_b}\right)^{2}\left(\frac{A^3}{h_b}\right)^{1/2}\right\} + 0.65 \quad (7)$$

where:

h_b → depth of the seabed at the location of the reef
H_o → significant wave height in deep water
s_B → depth of the reef crest
L_B → length of the submerged structure
A → dean scale parameter.

As demonstrated by Aragonés et al. (2015), major errors in the equilibrium profile can be committed, if the local factor of the data is disregarded. Therefore, in our case, we used the parameter A obtained from the method developed by Aragonés et al. (2015), which was based on the adjustment of real profiles in the study area. The result of the parameter for the study area was 0.0978.

Thus, we decided to apply Eqs. (2)–(7), in order to determine the characteristics of the reef, so that the conditions would be met for both studies (Black and Andrews, 2001a, 2001b; Ranasinghe et al., 2010), to be able to produce accretion on the beach. Thus, establishing the value of B/S as about 1, we obtained the width of the accretion of the offshore area as 63.4 m. Following the introduction of MFAR, the beach's width was 33.2 m at its maximum, and the long beach that was affected by accretion 270 m. Fig. 7 shows the MFAR within the area of the accretion cases according Ranasinghe et al. (2010), and the morphology of the hemitombolo created.

Moreover, the skill level of the surfers, is directly related to the kind of waves that would allow them to ride safely without jeopardizing their safety. The most important parameters, such as peel angle, and

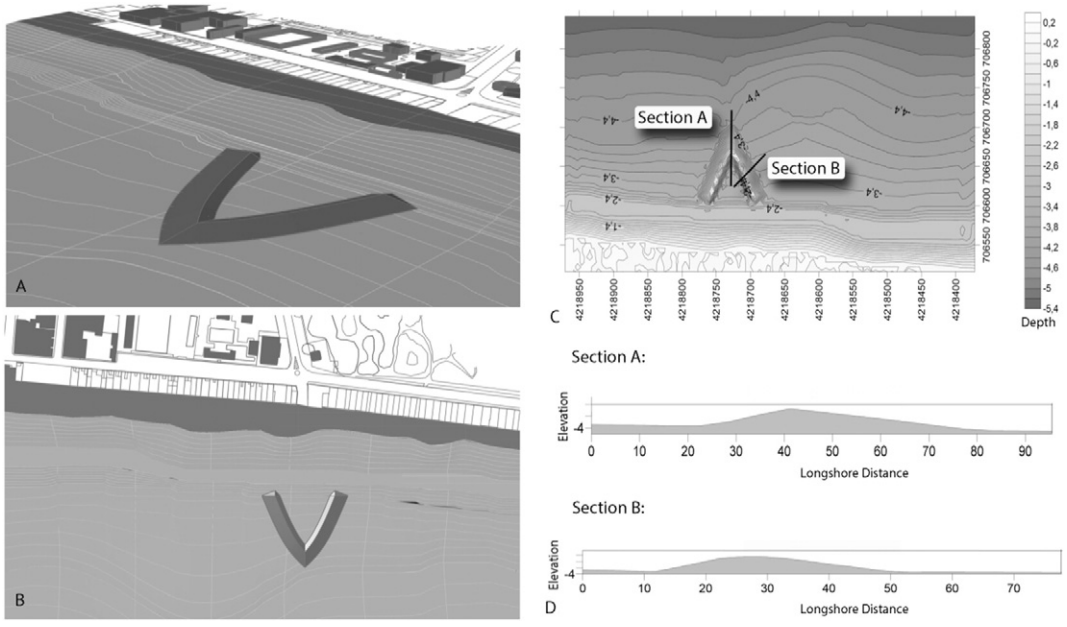


Fig. 8. A) and B) MFAR 3D model. C) Bathymetric model. D) Cross sections.

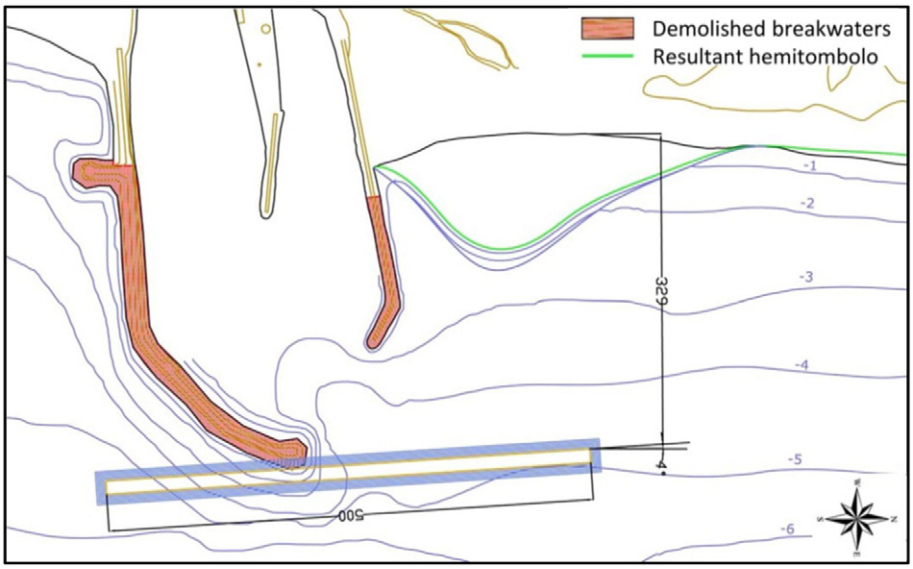


Fig. 9. Simulation of hemitombolo generated by the implementation of the dike and demolition of the breakwaters.

the classification of the quality of a wave from the standpoint of surfing, have been summarily described by Scarfe et al. (2003).

The morphology of the MFAR was designed so that the waves generated would be suited to the characteristics of the surfer in the area (Mediterranean surfer), which encompasses a range of skill levels that are between low and intermediate (between 1 and 6 ranking by Hutt et al. (2001)). Thus, based on studies by Henriquez (2004), West (2002) and Ten Voorde et al. (2006), with respect to the shape of wave breaking, we decided to focus on the reef in an abrupt way, and we estimated 66° as the angle between the vector of swell within the orthogonal and the arm of the reef, in order to encourage the concentration of the wave energy, and thereby, achieve a progressive wave breaking and wave control.

In order to reduce the cost of the reef, the nucleus obtaining a "V" shape was emptied, and it did not have any negative influence on the generated waves, according to the research of Aguiar et al. (2009).

Based on the above provisions, the designed MFAR can be distinguished in Fig. 8.

Main parameters and dimensions of the designed MFAR are summarized in the following points:

- Dimension of reef in orthogonal direction to the shore: 60 m
- Distance of breaking zone from the coast: ≈100 m
- Top offshore width of reef: 63.4 m
- Maximum ride generated: ≈65 m
- Internal angle of reef: 24°
- Water depth in the area of location: −4 m
- Reef crown depth: −0.7 m
- Offshore slope of reef: 1/10.

3.4. Modelling environment

Furthermore, for a hemitombolo to be generated there must be a supply of sand. The supply of sand in the area is being interrupted currently, by the existing breakwaters just off the mouth of the river (Fig.

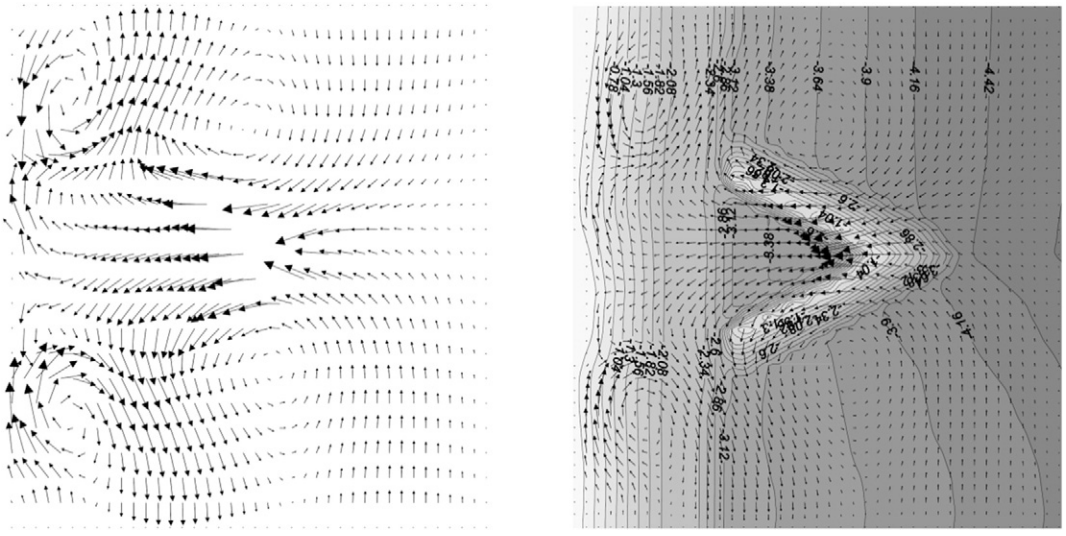


Fig. 10. System currents generated by the reef created before a wave normal incidence to it.

1B). That was why, we decided to replace two breakwaters by a dike to protect the mouth, and allow access to the marina. Besides, this dike was placed in such a way that it produces a hemitombolo, thereby, generating a bypass of sediment through the mouth of the river (Fig. 9).

According to data provided by the Marina, the minimum depth inside the port was 2.5 m–3 m. To maintain this depth, sufficient safety margin and a wide sufficient ground for the entry of vessels, we decided to locate the dike on the bathymetric −5 m, located at a distance of 329 m horizontally from the shoreline.

For the hemitombolo design, the length of the dike that was considered in the calculations, was the stretch of dike that faced the coast. The stretch of dike was opposite the mouth guard, and its only function was

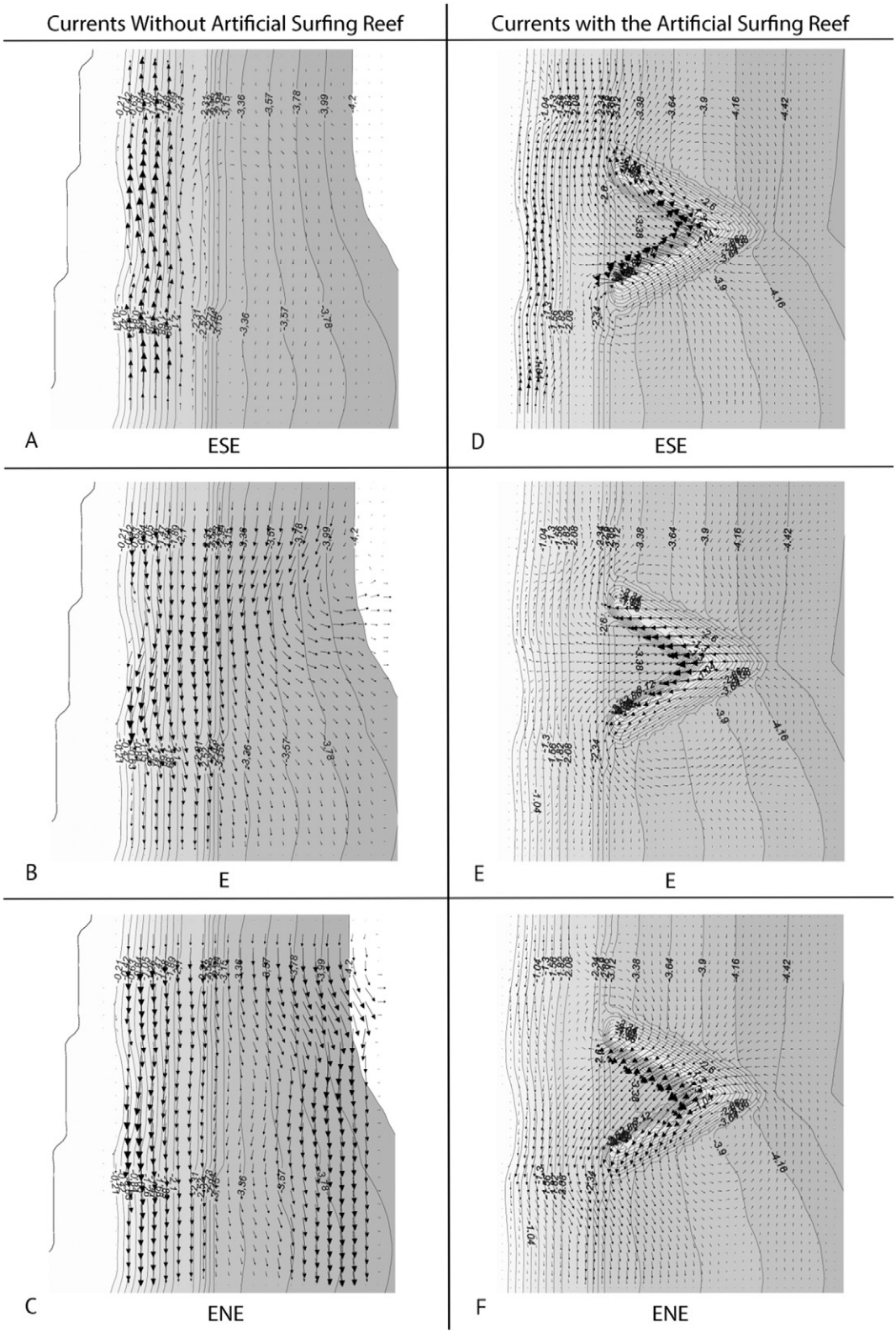


Fig. 11. Comparison of the regime of currents for the more frequently waves (ESE, E and ENE), in the case without MFAR (A, B and C) and with MFAR (D, E and F).

to create an area of calm waters at the mouth. The mouth had a width of about 260 m, and length was added to the designed dike. It was necessary that the tip of the hemitombolo be pronounced, so that it fulfilled the function of "by-pass" and the sediment did not flow into the mouth, therefore, following the same criteria as the MFAR (Black and Andrews, 2001a, 2001b; Ranasinghe et al., 2010), we agreed that the length of the dike must be 240 m, which will generate a hemitombolo of about 133 m (Fig. 9). Finally, the dike had an inclination of 4° with respect to the North, therefore, it was perpendicular to the mean flow.

3.5. SMC coastal modelling

The bathymetry was obtained from the study of the ECOLEVANTE (2006). The level of detail of the bathymetry was such that it was estimated that, on average, we would collect a total of 30–40 points/m^2 probe with a final accuracy of ±15 cm (ECOLEVANTE, 2006).

To make wave propagation for the area of study and to obtain the currents, the software of the Coastal Modelling System (www.smc.unican.es) was used, and it was similar to programs used in other studies such as the REF/DIF (Kirby and Özkan, 1994) used by Rakha and Abul-Azm (2000), and the LIMCIR developed by the Maritime Engineering Laboratory of the Polytechnic University of Catalonia (LIM-UPC), which was used by García Márquez (2009). These types of numerical models are based on a description of the most important physical processes, in relation to a given bathymetry, using a series of separate sub-models that simulate the main hydrodynamic mechanisms, such as wave propagation, currents generation, and so on. By unifying the technical criteria and the systematic organization of the numerical models, the reliability is increased in decision-making (González et al., 2007; González et al., 2004). The two-dimensional model of the current SMC program follows the Navier–Stokes equations.

The worst situation that a reef can face is that of a perpendicular incident wave, as it can produce a series of erosive currents in the shadow of it (Ranasinghe and Turner, 2006). In our study, the perpendicular waves of the area of study, was from the East and it was also the most frequent waves (Table 2). As it spread, a spectral wave whose characteristics (it might have been a monochromatic wave) were of a type of sea wave, was 1 m high and 10 s of period perpendicular to MFAR. This was done to check if erosion occurred around the reef.

The system of current generated by the spectral wave is shown in Fig. 10. This current system obtained indicates the achievement of a system of cells that promotes the accretion in the sheltered area of the multifunctional artificial reef. The error committed by the program when simulating the currents is 0.02 m/s for speed and one degree for direction.

Once it was proven that it was not an erosive situation before the perpendicular waves, we proceeded to study the whole incident waves (Table 2), comparing the currents generated in the area prior to the execution of the MFAR, to the currents generated after the implantation of the MFAR (Fig. 11). While watching, the reef execution caused a radical change of currents in favour of forming a protrusion shoreline. In Fig. 11D, E and F we can see how the MFAR generates currents that reduce solid longitudinal transport, and changes the direction and the magnitude of the existing currents in the area.

On the other hand, to check that the proposed MFAR was appropriate for the skill level of the surfers in the area (Mediterranean surfer), the study was based on the classification of peel angles generated and compared with graphics developed by Hutt et al. (2001); Walker (1974). This proved that the characteristics of the waves generated by the MFAR had an impact of a generic wave of 1.5 meter and 8 second period in the NE, ENE, E, ESE and E wave directions. This wave was chosen because, it has been proven that regular waves incident on these elements are usually between 0.7 m and 1.5 m wave height and periods of 7–9 s, and this is similar to the typical Mediterranean waves. In our case, 40.8% of the incident waves had these characteristics, while 57.4% had lower wave heights of 0.7 m. The remaining 1.8% had wave heights of over 1.5 m (Fig. 3).

In measuring the peel angles, a graphics program known as the Pixelmator was employed, to analyse the propagation models obtained by SCM. Estimation of the peel angle was conducted by measuring the spatial variation experienced by the phase of the wavefront passing through MFAR (Fig. 12).

From the study of different directions of incident wave with varying wave heights, we extracted a range within which, the surf generated by the MFAR moved. This range is represented by the skill levels required by the surfer, to be able to surf the wave generated under any condition. Fig. 13, shows the range of skill levels required by surfers, for the designed MFAR, which varies from 1 to 6, and is suited to the characteristics of a Mediterranean surfer.

Finally, the operation of the dike situated opposite the mouth was found. Using the SMC program, each of the directions of the waves were propagated, and as shown in Fig. 14, sediment transport occurs from north to south between the dam and the mouth of the river without accumulation of sediments at the mouth or around the dam.

3.6. Evaluation by the FEM of the seabed's behaviour

In order to analyse the mechanical behaviour (displacements and stresses) of the seabed against the load from the MFAR, a simulation by the FEM with the commercial software PLAXIS 3D Foundation has been carried out, where a reproduction of the geometry of an area of the seabed and the MFAR has been modelled, including the water cover both the seabed and the reef crest.

The total area of the model (120 × 120 × 17m) was defined to disregard boundary conditions related with the MFAR initial dimensions. For the simulations, 15-node wedge elements were used, with 27,150 elements and 76,995 nodes, locating over the MFAR surface the water level corresponding to 0.7 m (reef crest) and 3.3 m (seabed) (Fig. 15).

Two models were performed for different types of loose sand forming the seabed (IGME, 1994, 2005; MOPU, 1989; MAGRAMA, 2007), modifying the Young's Modulus and Poisson's ratio (E_{min} = 10.5 MPa and E_{max} = 24 MPa, ν_{min} = 0.20 and ν_{max} = 0.40) (Villalaz, 2004).

The test results concluded that the maximum vertical displacements (settlements) that would appear at both models is $135 \cdot 10^{-3}$ m and $42 \cdot 10^{-3}$ m respectively for the minimum and maximum values (Fig. 16A and B); in the case of the vertical effective stress (at a depth of 2.6 m – seabed – for disregarding the soil liquefaction) the value is, in both cases of 27.86 kN/m^2 (Fig. 16C).

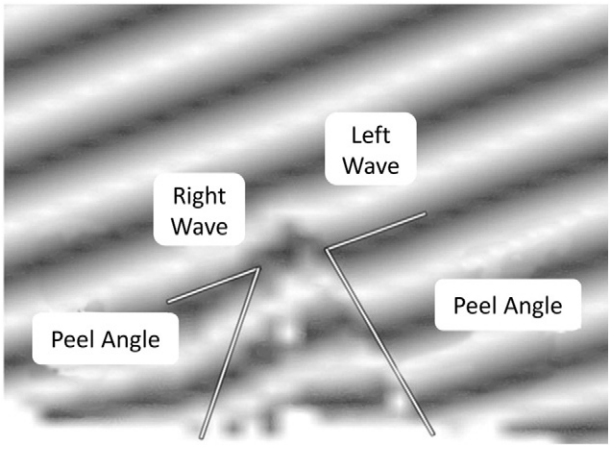


Fig. 12. Measuring peel angles for type waves.

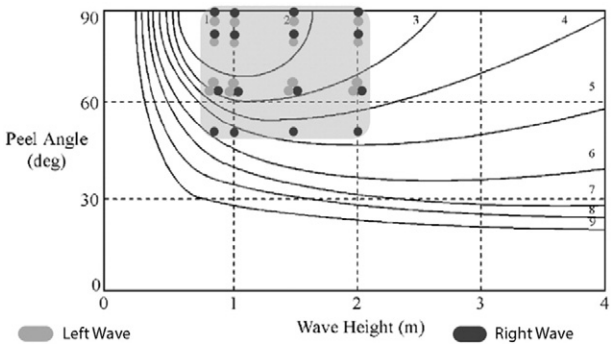


Fig. 13. Range of skill levels required in the designed MFAR. Modified from Hutt et al. (2001).

These values of settlements are consistent with simplified formulas of soil mechanics (GCOC, 2009) and according to Eq. (8),

$$S = \frac{V(1-\nu^2)}{1.25 \cdot E \cdot \sqrt{B \cdot L}} \tag{8}$$

where V is vertical load applied on the soil (due to the weight of MFAR), ν is the Poisson's ratio, E the Young's Modulus, with B and L as the width and the length of the MFAR respectively.

However, the values obtained can be increased 100% for taking into account the loose sand is sunk beneath the surface of the sea and its bearing capacity has been halved (Juárez and Rico, 1992; Terzaghi and Peck, 1948). Then, the final values of the vertical displacements would be:

$$S = \begin{cases} S_{min} = 120E{-}3m \\ \\ S_{max} = 46E{-}3m \end{cases} . \tag{9}$$

As can be observed both values are closed to the obtained with the FEM.

3.7. Reef structure, constructive process and cost analysis

According to Romero (1998), reefs that are used for the development and regeneration of the fauna marina, promoting reforestation, commissioning and reproduction of different species that colonize, must have the following characteristics: they must possess the greatest number of surfaces, holes and horizontal, vertical and inclined planes, that allow the entry of light, and streams, which are sufficient to enhance the productivity and colonization of species oxygenation.

As for the materials used to construct them, it has been observed that reefs that are built with rock or concrete, provide opportunities for habitat, but generally act as scavengers of fish rather than increase the number of finfish and diversity (Jackson et al., 2005). Meanwhile, geosynthetics provide a good substrate for a wide range of flora and fauna, and the extent and diversity of the habitat, allows adherence to marine, particularly plants in the shallower areas due to wave action (Jackson et al., 2004). Among the different geosynthetics, the one best suited for the construction of an artificial reef, is the non-woven geosynthetic, as it aids organisms growth, and the "soft" bodies (algae essentially) are mainly developed, while the type of geosynthetic that is woven, develops the "hard" bodies growth (molluscs, crustaceans, etc.) (Edwards, 2003). For his part, Jackson et al. (2005) compared the development of the ecosystem to three types of geosynthetics (polyester staple fibre needle punched non-woven, a composite dual layer mixed denier needle punched non-woven, and a sample of split film high strength polypropylene woven) in the Persian Gulf, and they observed that after 8 months of implementation, both non-woven geosynthetics provided good habitats for many species, thereby increasing diversity, and thus, being the composite layer geosynthetic, which provided the most diverse habitat.

Given the above, we decided to build the MFAR, by using geotextile containing a composite dual layer mixed denier needle punched non-woven (Fig. 17), and filled with sand, as these are the geosynthetics that have provided further development of the ecosystems in the areas which we used. Furthermore, the containers were positioned in such a way that, the gaps between them and enough surfaces for colonization, would favour the different species (Fig. 17).

The construction process chosen as optimal for execution has been executing MFAR in situ, using the anchorage and subsequent filling by pumping sand slurry. The process to follow is:

1. Geotextiles are manufactured with dual layer composite mixed denier needle punched non-woven on the beach with precise measurements, and are collected and assembled into a zone enabled for it.
2. A pontoon is loaded with all geotextile already manufactured and transported to its final location.
3. Using surveying equipment and divers, geotextiles are moored and anchored.

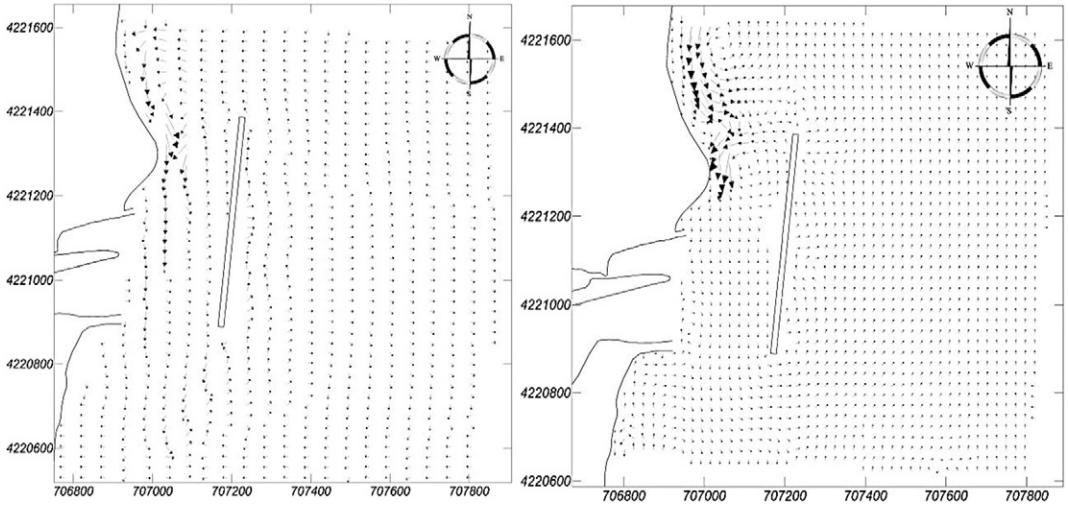


Fig. 14. Sediment transport at the mouth for directions ENE (right) and E (left).

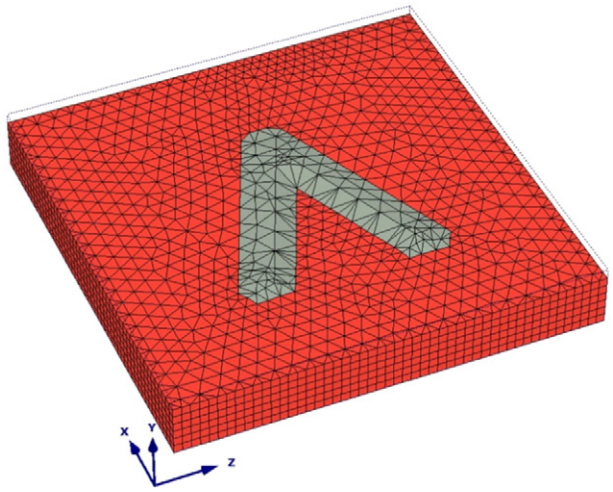


Fig. 15. MFAR model used in the simulation.

4. Geotextiles are filled by pumping sand and water slurry. Given the proximity of the reef to the shore, pumping will be done through a pumping station located on the beach.

MFAR volume designed is approximately 1920 m^3 (exactly 1919.28 m^3). Based on the cost per cubic meter of coral reefs made other executed with geotextiles (43 €/m^3 Narroweck, 285 €/m^3 Prattes, 243 €/m^3 Mount Maunganui, 193 €/m^3 Opunake, 332 €/m^3 Kovalam and 163 €/m^3), and the cost adapted to our area of study, an average cost of 210 €/m^3, a maximum cost of 332 €/m^3 and a minimum of 43 €/m^3 is assumed. So MFAR cost will be between 82,560 € and 637,440 €, being 403,200 € the most probable approximate average cost.

4. Discussion

When designing an anthropic element such as a MFAR, it is particularly important to study each and every one of the influential elements in it. The influential elements should be taken into account, and the different ecological design criteria and coastal engineering, should be combined. However, MFAR projects generally, tend to focus primarily on the numerical modelling of the reef and the system currents generated by the reef (Table 1). When designing a reef, failure to take into account the various elements and/or local variables may cause a negative result, as has happened to a number of submerged reefs due to poor orientation or an incorrect location (Ranasinghe and Turner, 2006) or the collapse of the structures in the sandy bottom due to settlements and liquefaction processes (Medina et al., 2006). It can be observed in parameter A of the formula of Ranasinghe et al. (2010), and indicated by Aragonés et al. (2015), that significant mistakes could be made, when the local factors in energy and sediment, are not taken into account. With this multidisciplinary work, a method for designing MFAR was proposed, which took into account, all local factors such as sediment transport, waves and currents, settlements, liquefaction, the skill level of surfers in the study area (so that the surf conditions generated by the reef, would suit their characteristics), the biocenosis, and the historical evolution of the area for proper location of the MFAR.

Apart from this, the location of such structures in the Mediterranean is backed by two main reasons: 1) In the Mediterranean, there is no readily available quality wave for surfing, so the possible recreational disuse of the structure is discarded, like what happened to the Prattes reef. 2) The microtide area, which will not affect the operation of the reef which can be used around 60% of the time, and would also prevent the structure from being visible occasionally, like what happened in other cases such as Boscombe.

From the study of the coastline and the biocenosis, the most suitable area for the location of the reef, was situated in the most degraded area in front of the houses of the Babilonia beach, due to the great loss of beach width, which occurred since the construction of the houses (1948), and had worsened in recent years, due to the wave reflection being able to reach the front of the houses (Fig. 5). On the other hand, Fig. 2C shows that the area covered by prairie *C. nodosa*, is much richer in species, than the community of properly calibrated sand, so placing the MFAR in this area between the height and line −6 coasts, will facilitate the development of algae and marine fauna, promoting diversity and richness of species in the area.

From the point of view of the coastal morphodynamics, by implementing the MFAR, we intended to make a change in the pre-existing currents, in order to prevent the tilting of the local beach, and then an accretion would occur on the beach. Such an accretion is generated in the shadow zone of the MFAR. Ranasinghe et al. (2010) established the need for four-cell currents to be generated. Of these four cells, the two closest to the coast, are those to produce a convergent flow toward the shore accretion, which is sought to be created on the beach. In Fig. 11E, as can be seen, the morphology developed by the MFAR was achieved, by the generation of these four cells, unlike what happened before the insertion of the MFAR, where the incident waves caused currents to be completely oriented to the south (Fig. 11B). On the other hand, for the beach accretion to occur, it was also necessary that there was a sediment supply, which is currently cut off by breakwaters that protect the mouth of the river and the marina (Fig. 6), as the current contribution of the River Segura is null (Aragonés et al., 2016), so the only contribution in the area is due to the dunes located from both north and south of the river. The supply of sand from the north accounted for about 315.3 m^3/year. To allow this sediment transport, we proposed replacing the breakwaters by dike, to protect the marina entrance, keep step with the entry draft for ships, at the same time generating a hemitombolo on the north of the beach mouth and create a bypass of sediments to the south (Fig. 14). However, the time it will take to produce the hemitombolo has not been calculated since this is conditioned by the actions that occur on the dunes, that is, over the dunes

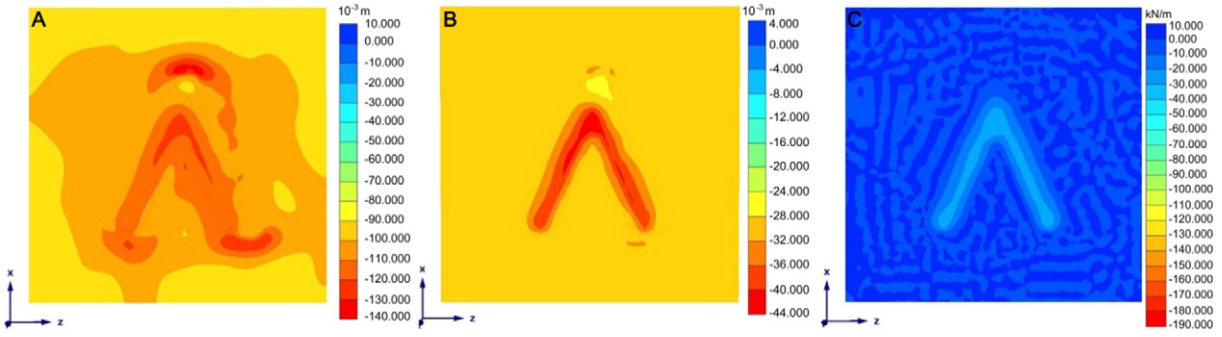


Fig. 16. A and B) Settlements from both models. C) Vertical effective stress.

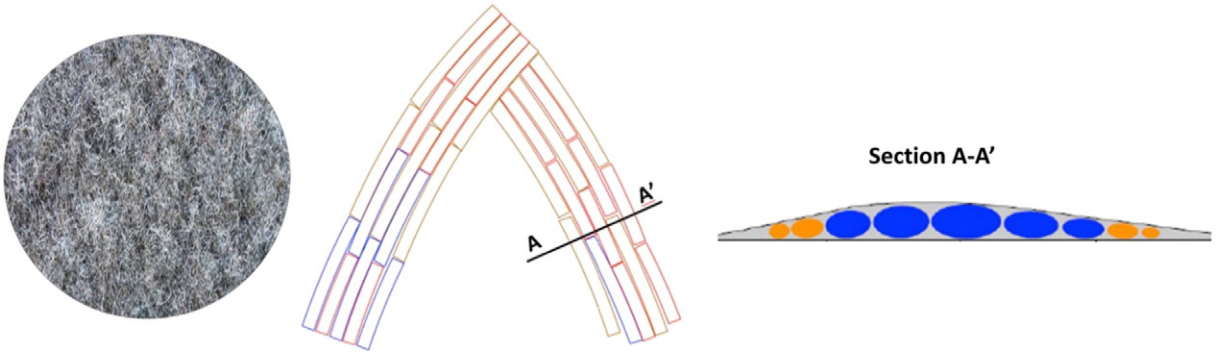


Fig. 17. Distribution of geotextile containers filled with sand and material employed.

are carrying out actions to stabilize, which can reduce sediment transport or even eliminate it entirely. Should the actions remove sediment transport completely, the hemitombolo be formed by artificial supply of sand.

In all cases regarding surfing, which are representative of wave conditions, they had an occurrence above 60% in the study area. Also, the projected structure promotes surfing, significantly improving the conditions of use, and adapting to the technical characteristics of the surfers in the area (Fig. 13). Therefore, as indicated by ASR (2008b) and Ng et al. (2014), surfing will be a source of income to the area, which will give added value to the investment.

Another important aspect to consider is the study of settlements and the soil liquefaction trying to avoid the mistakes that have been committed in other defence works beach (Medina et al., 2006). However, in the study area and 15 m thick sand, according to calculations made in the implementation area of the reef settlements are minimal (less than 13.5 cm; Fig. 16) and the liquefaction of the area can be ruled out. However, these are two important aspects to consider in this type of study.

The use of construction reef materials with similar characteristics to the natural environment lessened the impact of interventions by man, while incorporating features that increased the heterogeneity of the habitat, and improved the habitats of species targeted for ecological or economic interests. This amplified the value of the coastal structures, in relation to the natural marine ecosystem (Martins et al., 2010). Therefore, to promote the safety of the surfers, we decided to use geotextile containing a composite dual layer mixed denier needle punched nonwoven filled sand, which promotes the growth of "soft" bodies. Furthermore, the distribution of the geotextile was done to enable spaces between them, which facilitated the development of marine wildlife (Fig. 17). The construction will take place by anchoring empty geotextiles into place and the subsequent filling by pumping the slurry of water and sand from the beach. The average cost of building the MFAR is set in a 403,200 €.

5. Conclusion

The world's coasts are in a situation of continuous regression, particularly in Spain, where there are cases of missing dry beach meters per year. One such case is the Babilonia beach in Guardamar del Segura (Spain), in which the regression rate has been at about −0.69 m/year since 1992. This rate has increased dramatically in the last 5 years, due to reflection from the waves that directly affects the houses built in 1948. To alleviate this problem, a MFAR was designed with "V" morphology to work as part of the coastal defence, and a playful element to surfers. We have assumed that the method can be used by coastal engineers to design such structures.

Many studies have been conducted on the MFRA, however, there are few in which the coastal engineer can be supported for a full analysis of the processes to be followed in such studies. That is why in this article the climatic characteristics, biocenosis, sediment transport, settlements and liquefaction and the evolution of the coastline have been analysed. Concluding that for the design and construction of this type of work must take into account all the factors in this paper following the method mentioned in it.

The study of the evolution of the coastline suggested that in the south of the River Segura, the tilting shoreline is occurring in an anticlockwise direction. While in the north, it is leading to an accumulation of sediments in the north breakwater, which indicates that the longshore sediment transport is interrupted by these breakwaters. That is why we decided to replace them with a dike, in order to allow the flow of sediments.

Regarding the control of the currents derived from the implementation of MFAR, we generated an accretion with a maximum width of 33.2 m, which affected 270 m of studio beach, and this is almost the width of the dry beach in the study area, which was lost for a period of 49 years. On the other hand, the range of skills required by users of the wave generated revolves around beginners and intermediate levels, so that all profiled surfer types, can surf in the Mediterranean. However, it is suggested that this approach is accompanied by a scale study, like the one that was conducted for the Narrowneck reef by Turner et al. (2001), in order to verify the local effects that the formulations employed disregards.

Finally, we also took into account the biocenosis of the area, and placed the reef in an area of well calibrated fine sand, where diversity of species found was found to be lower than those in the area covered by prairie *C. nodosa*. This was done with the intention of fostering the development of the marine ecosystem, promoting diversity and species richness of the area.

Acknowledgement

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